# Mahiuddin Al Kamal

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## Work Experience

**Working Student, T-Systems Multimedia Solutions GmbH, Germany**
*February, 2021- Present*
**Working area**: Confidential Computing (*Security Architecture*)
**Responsibilities:**

- Researching and analyzing current available technologies and frameworks supporting Confidential Computing
- Migrating legacy applications to confidential computing enabled applications
- Developing confidential computing enabled applications from scratch

**Technologies using:**
Intel SGX, Intel SDK, Open Enclave SDK, SCONE, Graphene, Occlum, EGo, Marblerun, Decentriq, Azure Confidential Computing

**Research Intern, Systems Engineering Lab, Chair of Systems Engineering, TU Dresden, Germany**
*October, 2017- February, 2019*
**Responsibilities:**

- Researched and analyzed on ARM's Platform Security Architecture Attestation for IoT devices and Microsoft's Azure Attestation on Trusted Execution Environments such as Intel SGX Enclaves

**Assistant Software Engineer and Intern, Millennium Information Solution Limited, Bangladesh**
*October, 2017- February, 2019*
**Working department**: Sylvia (*Human Resource Management System Solution*)
**Responsibilities:**

- Developed different modules (*e.g.: Recruitment, Leave, Attendance, Payroll, Staff Benefit Management, Succession Management etc.*) for 'Sylvia'
- Developed a fully responsive e-Recruitment system for different renowned private banks of Bangladesh (*e.g.: UCBL, NRB Bank, AIBL etc.*)
- Developed different types of complex business reports for 'Sylvia' application using RDLC and Crystal Reporting Tool

**Technologies used:**
ASP.NET, C#, Oracle 11g, PostgreSQL, Microsoft Server 2012, Visual Studio, Toad, pgAdmin, Git, SVN

## Education

- **Master's in Distributed Systems and Engineering *(Oct,2019 - Present)***
  Technische Universität Dresden, Germany
  Major Courses: Systems Engineering, Transnational Information Systems, Mobile Communication and Mobile Computing, Distributed Systems, Security and Cryptography, Design Patterns and Frameworks etc.
  CGPA: 2.0 (*after third semester*)
- **B.Sc. in Computer Science and Engineering *(May,2012 - March,2017)***
  Chittagong University of Engineering and Technology (*CUET*), Chittagong, Bangladesh
  Major Courses: Data Structure, Algorithms, Object Oriented Programming, Discrete Mathematics, Systems Architecture, Compiler Design, Operating System, Artificial Intelligence etc.
  CGPA: 3.55 out of 4.00

Merit Position: 13th (*out of 112*)

- **Higher Secondary School Certificate *(June,2009 - July,2011)***
  Al-Amin Academy School and College, Chandpur, Bangladesh
  GPA: 5.00 out of 5.00
- **Secondary School Certificate *(January,2007 - May,2009)***
  Al-Amin Academy School and College, Chandpur, Bangladesh
  GPA: 5.00 out of 5.00 (A+ in all subjects)

## Projects

***Vehicle Number Plate Detection and Recognition in Bengali***

It's a complete system to detect and extract the plate in a vehicle image and recognize the numbers or characters from the plate for Bangladeshi Vehicles. It is based on my bachelor's thesis named '*Vehicle License Plate Detection and Recognition Based on Kirsch Algorithm and Hybrid Proximal Approach*' supervised by Prof. Dr. Pranab Kumar Dhar. A simple algorithm for vehicle license plate detection and recognition for Bangladeshi vehicles is proposed in this thesis. Here, the edges are detected using Kirsch Algorithm. A combined edge histogram value analysis & mathematical Morphology based approach is used to determine the license plate region. After that, Hybrid Proximal Approach is used to correct the skew angle of the image if available. Then by using Connected Component Labeling, Bounding Box and Histogram value the characters and digits are segmented. Finally, Template Matching is used to recognize the characters and the digits of the Bangla license plate. The proposed system works smoothly for images taken under various conditions.

*Research Area: Image Processing.*

*Tool: MATLAB.*

***MobiCon***

It's an android app which is used to ease our finding of latest smartphones available in the country. One can use this app to know the specifications & latest price in the local market, can give reviews and ratings, can also search the nearest location one can find to avail that smartphone and many more features such as - reviews, comparison etc. Here, we used PHP for backend operations and MySQL for database. With this app, we completed our '*Android Mobile Apps Development Trainer*' training program conducted by the Ministry of ICT, Bangladesh.

*Tools: Android, Java, PHP, MySQL, GPS, Eclipse ADT.*

***Easy Transport***

It's an IoT Project which can implement a fully automated transport system in respective to our locale transport (e.g.: Bus, Train etc.) using RFID card without actual cash transaction. Here, an android app is used to manage the Arduino module. Passengers use their individual RFID card to travel from one place to another by just swiping the card near to the Arduino module which is attached to the Bus/Train door. The cards can be recharged in the recharge points. Thus, one can move anywhere any time using the card only without any actual cash in hand.

*Tools: Arduino IDE, Android Studio, FireBase, RFID card*

***Ghorowa E-shop***

It's an e-commerce website for online shopping. It is fully responsive and can be used for any kind of small online business operation. Here, PHP is used for backend and MySQL is used as a database. HTML, CSS, Bootstrap is used for front-end.

*Tools: HTML, CSS, Bootstrap, PHP, MySQL.*

## Technical Skills

***Programming Language*:** C#, Python, Java, C++, C, MATLAB, LaTeX
***Database*:** Oracle 11g, PostgreSQL, MySQL
***Server:*** Microsoft Windows Server 2012/2016
***Version Control System*:** Git, SVN
***Operating System:*** Windows, Linux (*Ubuntu*)

***Tools*:** Docker, Kubernetes, Microsoft Azure, VS Code, PyCharm, Visual Studio, Eclipse, NetBeans, Toad, pgAdmin, phpMyAdmin, ProVerif, PRISM

## Trainings

**ASP.NET Developer Training**

Institute: Ernst & Young (*Funded by Ministry of ICT, Bangladesh*)
Topics Covered: OOP concepts, Basic and Advanced C# concepts, ASP.NET (*Web forms, MVC, Windows form application*) and so on.
Duration: 4 months (*Nov, 2016 – Feb, 2017*)
Location: Chittagong University of Engineering and Technology, Chittagong, Bangladesh

**Android Mobile Apps Development Trainer**

Institute: Ethics Advanced Technology Limited (*Funded by Ministry of ICT, Bangladesh*)
Topics Covered: Basic concepts of Java & Advanced concepts of Android and so on.
Duration: 6 months (*Jan, 2016- June, 2016*)
Location: Chittagong University of Engineering and Technology, Chittagong, Bangladesh

## Awards & Certificates

- 1st Runner-up in the 'National Mobile App Development Training' conducted by Ministry of ICT (*App Name* - ***MOBICON***).
- Certificate on 'Android Mobile Apps Development Trainer' upon a successful completion of the training program conducted by the Ministry of ICT, Bangladesh.
- University Merit Scholarship.
- Honorable mention for a science project named 'Industrial Production of Urea from Urine' in the '31st National Science and Information & Communication Technology Science Fair' conducted by Ministry of Education, Bangladesh in 2010.
- Secondary School Merit Scholarship (*General grade*) by Ministry of Education, Bangladesh in 2009.
- The Students Welfare Merit Scholarship (*Grade - A*) by The Students Welfare Association, Chandpur, Bangladesh in 2009.
- Junior School Merit Scholarship (*General grade - 1st in Upazilla*) by Ministry of Education, Bangladesh in 2006.
- 1st position (*as a team*) in the General Knowledge Competition conducted by the 'Bangladesh Shishu Academy' in Sylhet, Bangladesh in 2006.
- Town Club Merit Foundation Scholarship (*Ranked 2nd*) by Nahar-Amin Merit Foundation, Sylhet, Bangladesh in 2003.
- 2nd prize for two categories (*General Knowledge & Hand Writing*) in a competition conducted by a local cultural organization named 'Shopneel Renesa' at their foundation anniversary in Sylhet, Bangladesh in 2005.
- Primary School Merit Scholarship (*Talent pool - 2nd in Upazilla*) by Ministry of Education, Bangladesh in 2003.
- Nahar Amin Merit Scholarship (*Grade - A*) by Nahar-Amin Merit Foundation, Sylhet, Bangladesh in 2003.
- 1st Prize in 'Essay Writing' in a Talent Hunt Competition conducted by Bangladesh SCOUTS, Sylhet, Bangladesh in 2003.

## Language Skills

Fluent in English (*both written and verbally*), IELTS (*CEFR-C1*), Basic German (*CEFR-A1*), Bengali (*Native*)

## Memberships

Bishwo Shahitto Kendro (*2007-2011*)
Bangladesh SCOUTS (*2003-2006*)

## Voluntary Experiences

- Organized a five day long free 'Eye Treatment & Lens Transplant Camp' as a Bangladesh SCOUTS member in the year 2006 and 2008 in Kulaura, Moulavibazar, Sylhet, Bangladesh.
- Distributed emergency foods among the poor flood-affected people in a devastating flood occurred in 2004.

## Personal Traits

Optimist, Passionate learner, Dreamer, Believer, Cheerful, Confident, Tech enthusiast

## Hobbies

Exploring Mountains and Historical Places, Watching Tech Videos, Watching Movies/Drama Series, Playing Games